

# Advanced Algorithms Lab

## Lab-4: Shortest Path using Dijkstra's Algorithm

### Aim

To find the shortest path in a weighted graph using Dijkstra's algorithm.

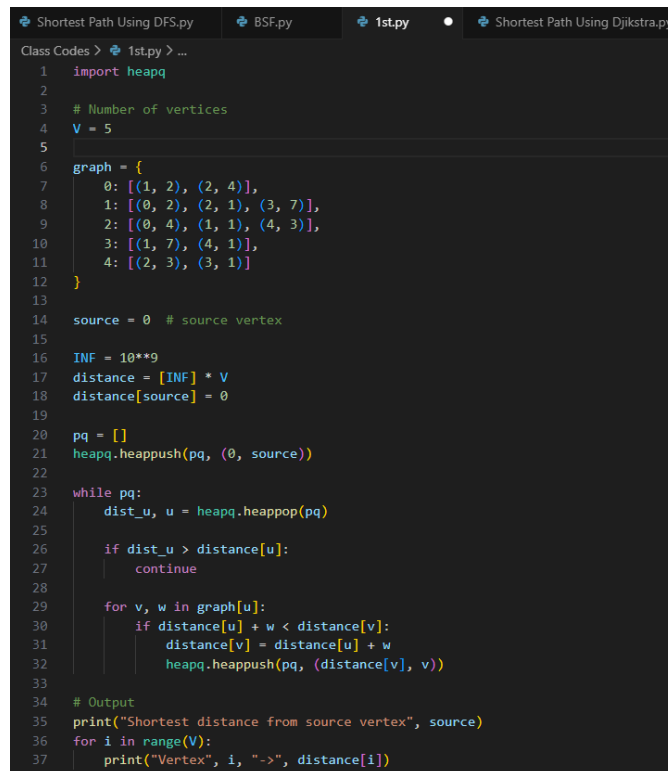
### Theory

Dijkstra's algorithm calculates the shortest distance from a source vertex to all other vertices in a graph with non-negative edge weights.

### Algorithm (Brief)

- Initialize all distances as infinity.
- Set source distance to zero.
- Use priority queue to select minimum distance vertex.
- Relax all adjacent edges.

### Code Screenshot

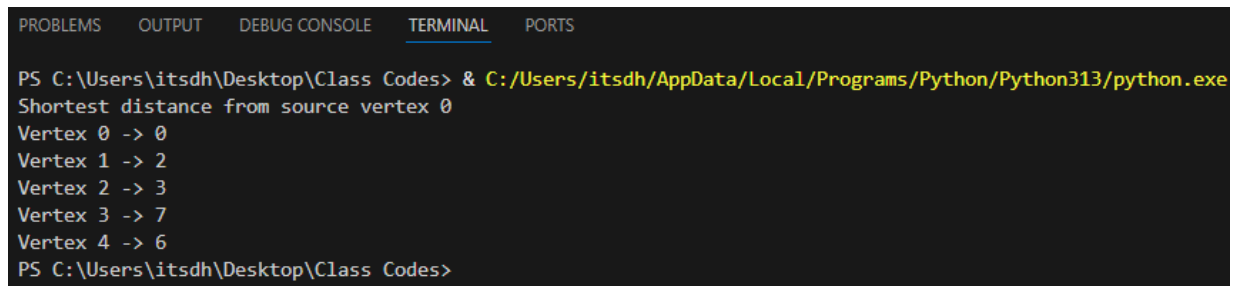


```
Shortest Path Using DFS.py  BFS.py  1st.py  Shortest Path Using Dijkstra.py
Class Codes > 1st.py > ...
1  import heapq
2
3  # Number of vertices
4  V = 5
5
6  graph = {
7      0: [(1, 2), (2, 4)],
8      1: [(0, 2), (2, 1), (3, 7)],
9      2: [(0, 4), (1, 1), (4, 3)],
10     3: [(1, 7), (4, 1)],
11     4: [(2, 3), (3, 1)]
12 }
13
14 source = 0 # source vertex
15
16 INF = 10**9
17 distance = [INF] * V
18 distance[source] = 0
19
20 pq = []
21 heapq.heappush(pq, (0, source))
22
23 while pq:
24     dist_u, u = heapq.heappop(pq)
25
26     if dist_u > distance[u]:
27         continue
28
29     for v, w in graph[u]:
30         if distance[u] + w < distance[v]:
31             distance[v] = distance[u] + w
32             heapq.heappush(pq, (distance[v], v))
33
34 # Output
35 print("Shortest distance from source vertex", source)
36 for i in range(V):
37     print("Vertex", i, "->", distance[i])
```

## Input

- Number of vertices
- Number of edges
- Weighted edge list
- Source vertex

## Output Screenshot



```
PROBLEMS  OUTPUT  DEBUG CONSOLE  TERMINAL  PORTS

PS C:\Users\itsdh\Desktop\Class Codes> & C:/Users/itsdh/AppData/Local/Programs/Python/Python313/python.exe
Shortest distance from source vertex 0
Vertex 0 -> 0
Vertex 1 -> 2
Vertex 2 -> 3
Vertex 3 -> 7
Vertex 4 -> 6
PS C:\Users\itsdh\Desktop\Class Codes>
```

## Time Complexity

$$O((V + E) \log V)$$

## Result

Thus, the shortest path in a weighted graph was successfully found using Dijkstra's algorithm.